

B.E. Electrical Engineering Second Year Second Semester Examination 2024

ELECTRICAL INSTRUMENTATION PART – II

Examination Solutions

Solution to Question 1. (a)

Problem Statement: Elucidate with a diagram, the brief details of functional blocks of a Cathode Ray Oscilloscope (CRO).

Answer

A **Cathode Ray Oscilloscope (CRO)** is an electronic measuring instrument used to display and measure voltage signals as a function of time. It is commonly used for measuring amplitude, time period, frequency and phase difference of electrical signals.

The CRO converts an electrical signal into a visible waveform on the screen of a cathode ray tube. The vertical deflection of the spot represents the instantaneous value of the input voltage, while the horizontal deflection represents time.

1. Functional Block Diagram of CRO

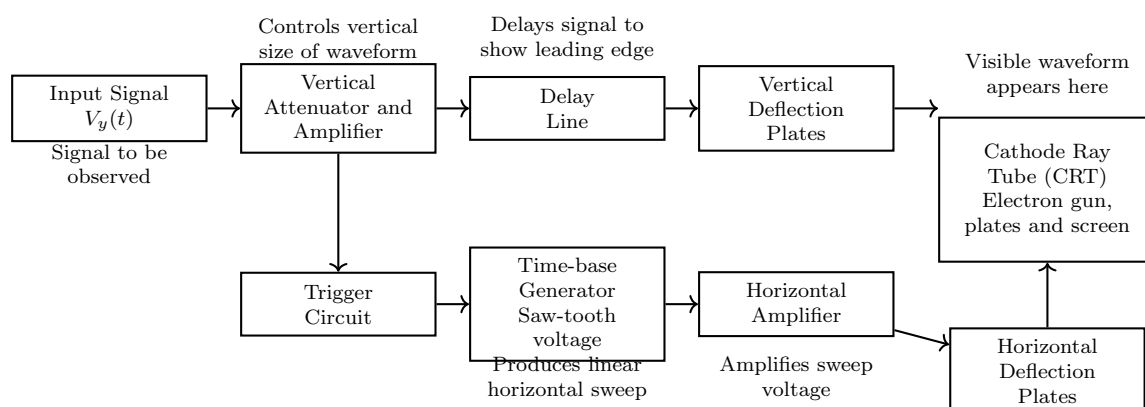


Figure 1: Functional block diagram of a Cathode Ray Oscilloscope.

2. Brief Details of Functional Blocks

1. Cathode Ray Tube (CRT):

The CRT is the heart of the CRO. It produces a narrow beam of electrons, accelerates the beam, focuses it and finally allows it to strike the fluorescent screen. When

the electron beam strikes the screen, a luminous spot is produced. By moving this spot horizontally and vertically, the waveform is displayed.

The CRT mainly contains:

electron gun, deflection plates and fluorescent screen.

2. Electron Gun:

The electron gun produces the electron beam. It consists of heater, cathode, control grid, focusing anode and accelerating anodes. The cathode emits electrons when heated. The control grid controls the intensity of the beam. The focusing anode focuses the beam into a fine spot, and the accelerating anode gives high velocity to the beam.

3. Vertical Deflection System:

The signal to be observed is applied to the vertical input terminals. This signal is passed through an attenuator and a vertical amplifier. The vertical amplifier increases the signal level so that it can produce sufficient vertical deflection on the screen.

The vertical deflection represents the instantaneous value of the input voltage.

4. Delay Line:

The delay line is inserted in the vertical channel before the signal reaches the vertical deflection plates. The trigger and time-base circuits require a small time to start the horizontal sweep. Due to this delay, the beginning or leading edge of the input waveform can be properly observed on the screen.

5. Trigger Circuit:

The trigger circuit synchronises the horizontal sweep with the input signal. It ensures that every sweep starts at the same point of the waveform. Hence, the waveform appears steady and stationary on the screen.

6. Time-base Generator:

The time-base generator produces a saw-tooth voltage. This voltage is applied to the horizontal deflection system. During the rising part of the saw-tooth waveform, the electron beam moves linearly from left to right. Thus, the horizontal axis of the CRO represents time.

7. Horizontal Amplifier:

The horizontal amplifier amplifies the saw-tooth sweep voltage before applying it to the horizontal deflection plates. This controls the horizontal movement of the electron beam.

8. Deflection Plates:

The CRO has two pairs of deflection plates:

vertical deflection plates and horizontal deflection plates.

The vertical plates move the electron beam up and down. The horizontal plates move the electron beam left and right. By the simultaneous action of both deflection systems, the waveform is drawn on the screen.

9. Fluorescent Screen:

The screen is coated with fluorescent material. When the high-speed electron beam strikes the screen, a bright spot is produced. The movement of this spot forms the waveform.

10. Power Supply:

The CRO requires low-voltage and high-voltage power supplies. Low-voltage supplies are used for amplifiers and control circuits, while high-voltage supplies are used for accelerating and focusing the electron beam.

3. Simple Internal Diagram of CRT

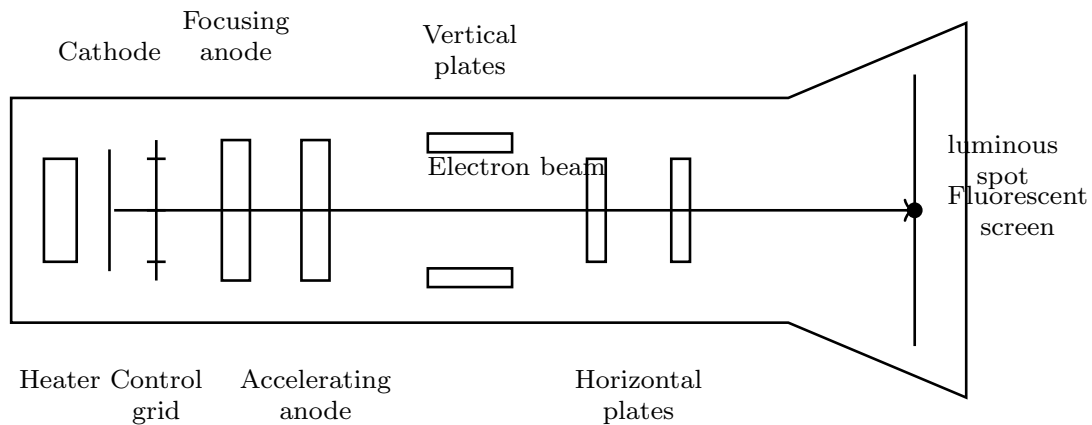


Figure 2: Simplified internal diagram of Cathode Ray Tube.

Final Answer

Thus, the CRO consists of several important functional blocks such as CRT, vertical amplifier, delay line, trigger circuit, time-base generator, horizontal amplifier, deflection plates and power supply. These blocks together convert the applied electrical signal into a visible waveform on the fluorescent screen.

Solution to Question 1. (b)

Problem Statement: How can the phase difference between two sinusoidal signals having same frequency be measured with the help of CRO?

Answer

The phase difference between two sinusoidal signals having the same frequency can be measured with the help of a CRO by using the **X-Y mode**. In this method, the internal time-base of the CRO is switched off.

One sinusoidal signal is applied to the horizontal input, i.e. X-input, and the other sinusoidal signal is applied to the vertical input, i.e. Y-input. Since both signals have the same frequency, a stationary Lissajous figure is formed on the CRO screen.

1. Applied Signals

Let the signal applied to the X-input be

$$x = x_0 \sin \omega t$$

and the signal applied to the Y-input be

$$y = y_0 \sin(\omega t + \phi)$$

where,

ϕ = phase difference between the two sinusoidal signals.

2. Lissajous Figure for Same Frequency Signals

When two sinusoidal signals of the same frequency are applied to the X and Y inputs of the CRO, the Lissajous pattern becomes an ellipse. The phase angle can be calculated from the intercepts of this ellipse.

3. Derivation of Formula

Given,

$$x = x_0 \sin \omega t$$

$$y = y_0 \sin(\omega t + \phi)$$

Expanding the second equation,

$$y = y_0 [\sin \omega t \cos \phi + \cos \omega t \sin \phi]$$

At the instant when the ellipse cuts the Y-axis,

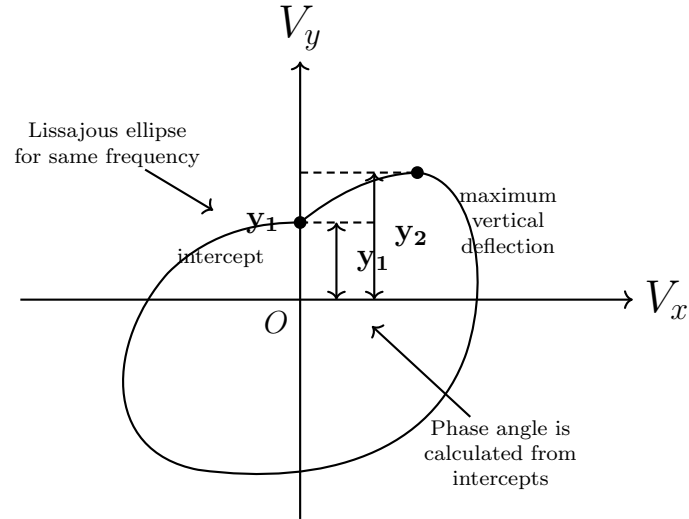


Figure 3: Lissajous ellipse for two sinusoidal signals of the same frequency.

$$x = 0$$

Therefore,

$$x_0 \sin \omega t = 0$$

$$\sin \omega t = 0$$

So,

$$\omega t = 0, \pi, 2\pi, \dots$$

At this instant,

$$\cos \omega t = \pm 1$$

Therefore,

$$y = y_0[0 \cdot \cos \phi + \cos \omega t \sin \phi]$$

$$y = \pm y_0 \sin \phi$$

Taking magnitude,

$$y_1 = y_0 \sin \phi$$

where y_1 is the intercept of the ellipse on the Y-axis.

The maximum vertical deflection is

$$y_2 = y_0$$

Therefore,

$$\frac{y_1}{y_2} = \frac{y_0 \sin \phi}{y_0}$$

$$\frac{y_1}{y_2} = \sin \phi$$

Hence,

$$\sin \phi = \frac{y_1}{y_2}$$

and

$$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$$

4. Alternative Formula Using X-axis

Similarly, if x_1 is the intercept on the X-axis and x_2 is the maximum horizontal deflection, then

$$\sin \phi = \frac{x_1}{x_2}$$

and

$$\phi = \sin^{-1} \left(\frac{x_1}{x_2} \right)$$

5. Special Cases

$$\phi = 0^\circ$$

The Lissajous figure becomes a straight line with positive slope.

$$\phi = 90^\circ$$

The Lissajous figure becomes a circle if the two signal amplitudes are equal.

$$\phi = 180^\circ$$

The Lissajous figure becomes a straight line with negative slope.

6. Procedure

1. Switch the CRO to X-Y mode.
2. Apply one sinusoidal signal to the X-input.
3. Apply the other sinusoidal signal to the Y-input.
4. Adjust the gain controls so that a clear stationary ellipse is obtained.

5. Measure the intercept y_1 on the Y-axis.
6. Measure the maximum vertical deflection y_2 .
7. Calculate the phase difference using

$$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right).$$

Final Answer

Thus, the phase difference between two sinusoidal signals of the same frequency can be measured by applying them to the X and Y inputs of the CRO and observing the Lissajous figure.

$$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$$

or,

$$\phi = \sin^{-1} \left(\frac{x_1}{x_2} \right)$$

Solution to Question 1. (OR) (a)

Problem Statement: Why is a steady horizontal plate sweep voltage in saw-tooth wave shape necessary in the function of the CRO? Draw and explain the basic operation of such wave generating circuit.

Answer

In a Cathode Ray Oscilloscope (CRO), the input signal is applied to the vertical deflection plates. Therefore, the vertical movement of the electron beam represents the instantaneous value of the input voltage.

To display the variation of this signal with time, the electron beam must also move horizontally from left to right with uniform speed. For this purpose, a **saw-tooth sweep voltage** is applied to the horizontal deflection plates.

1. Necessity of Saw-tooth Sweep Voltage

A saw-tooth voltage has two parts:

linear rising part and fast falling part

During the linear rising part, the horizontal deflection voltage increases uniformly with time.

$$V_x(t) = kt$$

where k is a constant.

Since the horizontal deflection of the electron beam is proportional to the applied horizontal voltage,

$$x \propto V_x(t)$$

Therefore,

$$x \propto kt$$

Hence,

$$x \propto t$$

So, the horizontal axis of the CRO screen becomes a time axis.

2. Why Linear Sweep is Necessary

1. **Uniform movement of spot:** The rising portion of the saw-tooth voltage makes the electron beam move from left to right with constant speed.
2. **Time representation:** Since $V_x(t) = kt$, equal horizontal distances on the screen represent equal intervals of time.

3. **Correct waveform shape:** If the sweep voltage is not linear, the horizontal motion will not be uniform. Then the waveform will be compressed in some regions and expanded in other regions.
4. **Fast retrace:** The sudden fall of the saw-tooth voltage brings the electron beam quickly back to the left side of the screen for the next sweep.
5. **Stable display of periodic signal:** With proper triggering, each sweep starts from the same point of the input waveform. Hence, a stable waveform is obtained on the screen.

3. Waveform of Saw-tooth Sweep Voltage

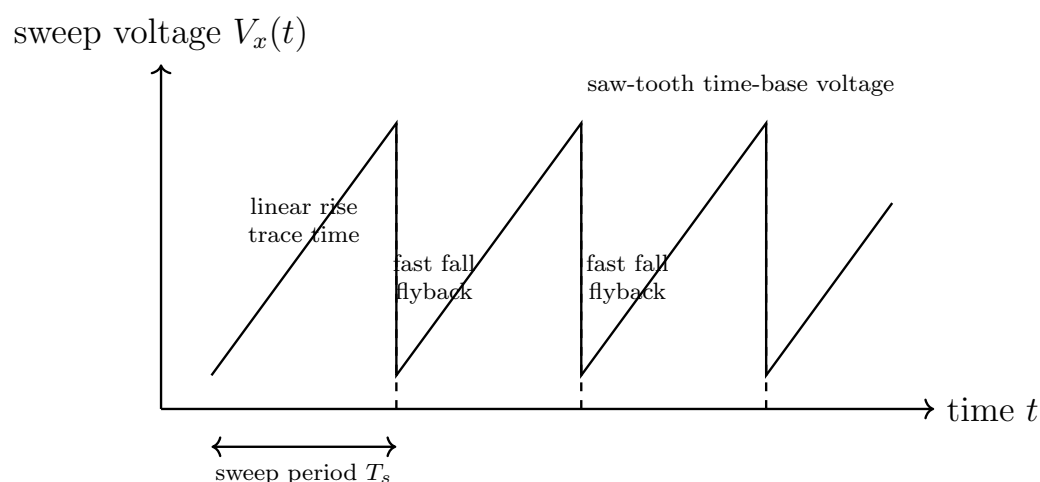


Figure 4: Saw-tooth sweep voltage used for horizontal deflection in CRO.

4. Basic Time-base or Sweep Generating Circuit

A simple sweep generator may be realised using an RC charging circuit and a discharge switch. The capacitor charges slowly through a resistor and discharges rapidly through a switching device. The voltage across the capacitor is the saw-tooth sweep voltage.

5. Operation of the Sweep Generator

1. **Charging interval:** When the discharge switch is open, the capacitor C charges through the resistor R . The capacitor voltage rises approximately linearly over the useful part of charging.

$$V_C(t) \approx kt$$

This voltage is applied to the horizontal deflection system.

2. **Trace interval:** During charging, the voltage applied to the horizontal plates increases steadily. Hence, the electron beam moves from left to right with nearly uniform speed. This is called the trace or sweep interval.

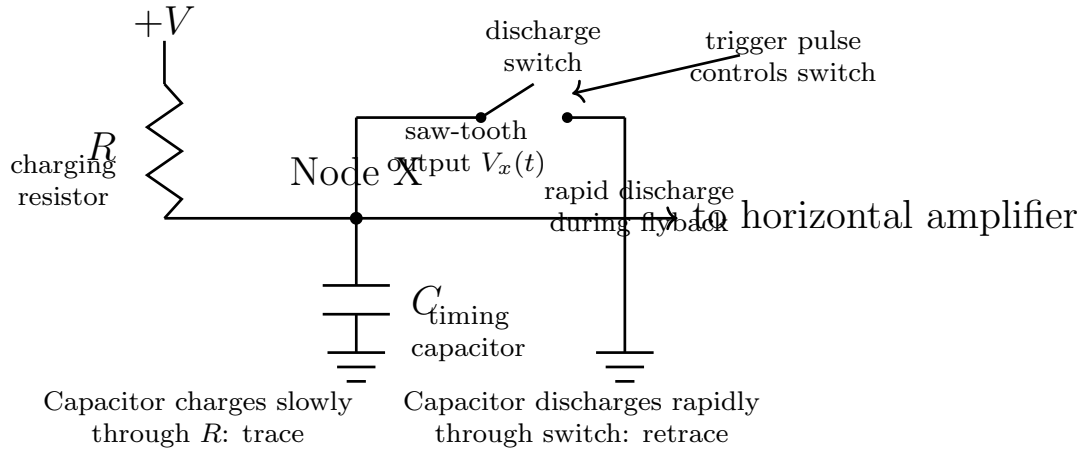


Figure 5: Basic saw-tooth sweep voltage generating circuit.

3. **Discharge interval:** At the end of the sweep, a trigger pulse closes the discharge switch. The capacitor discharges very rapidly through the switch.
4. **Flyback interval:** During rapid discharge, the horizontal voltage falls suddenly. Hence, the electron beam returns quickly to the left side of the screen. This is called flyback or retrace.
5. **Repetition:** After discharge, the switch opens again and the capacitor starts charging once more. Thus, a repetitive saw-tooth voltage is obtained.

6. Final Answer

Thus, a steady saw-tooth sweep voltage is necessary in a CRO because it makes the horizontal deflection proportional to time. The linear rising part gives uniform left-to-right motion of the spot, and the rapid falling part brings the spot back quickly for the next sweep.

$$V_x(t) = kt$$

Therefore, the CRO screen displays the input voltage waveform as a function of time.

Solution to Question 1. (OR) (b)

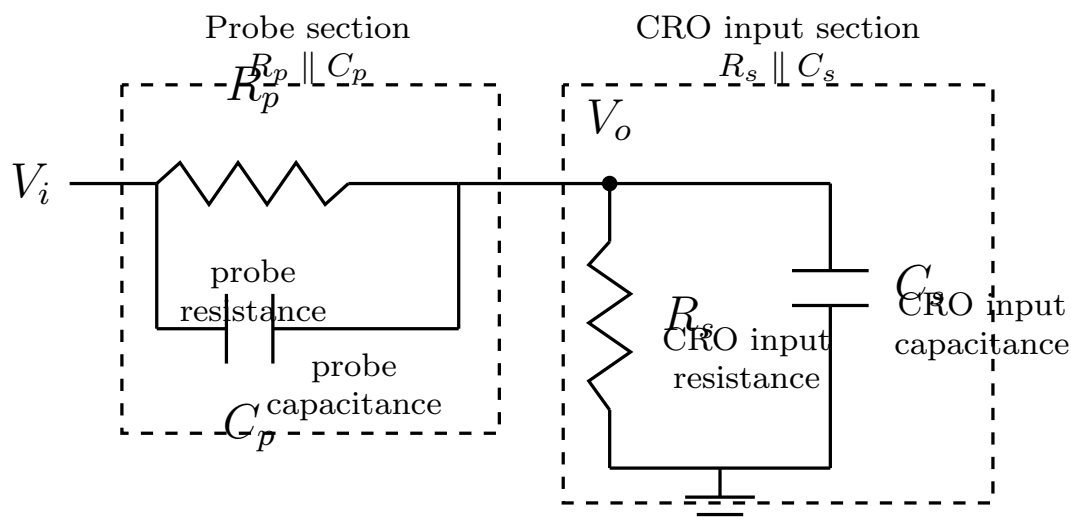
Problem Statement: Explain the necessity of matched probe for a CRO.

Answer

A CRO probe is used to connect the signal source to the oscilloscope input. In practice, the CRO input is not purely resistive. It has an input resistance and an input capacitance.

Similarly, the probe also has resistance and capacitance. Therefore, the probe and the CRO input form a frequency-dependent voltage divider. If the probe is not properly matched or compensated, different frequency components of the signal will be attenuated by different amounts. As a result, the displayed waveform becomes distorted.

1. Equivalent Circuit of CRO Probe and Input



$$\text{For matched probe: } R_p C_p = R_s C_s$$

Figure 6: Equivalent circuit of a matched CRO probe connected to the CRO input.

2. Condition for Proper Matching

Let,

$$R_p = \text{probe resistance}$$

$$C_p = \text{probe capacitance}$$

$$R_s = \text{CRO input resistance}$$

C_s = CRO input capacitance

For proper compensation or matching, the time constant of the probe section must be equal to the time constant of the CRO input section.

$$R_p C_p = R_s C_s$$

This condition makes the attenuation independent of frequency.

3. Necessity of Matched Probe

1. **To avoid waveform distortion:** A CRO input has capacitance. At high frequency, capacitive reactance changes. If the probe is not matched, high-frequency components are attenuated differently from low-frequency components. Therefore, the waveform shape becomes distorted.
2. **To obtain frequency-independent attenuation:** A matched probe behaves like a compensated voltage divider. Its attenuation remains almost constant over a wide range of frequencies.
3. **To reduce loading effect:** If the CRO is connected directly to a circuit, it may draw current and change the actual circuit voltage. A probe increases the effective input impedance and reduces loading on the circuit under test.
4. **To display fast-changing signals correctly:** Square waves, pulses and high-frequency signals contain high-frequency components. A mismatched probe may round the edges or produce overshoot. A matched probe shows the waveform more faithfully.
5. **To improve measurement accuracy:** Proper matching helps in accurate measurement of amplitude, rise time, time period and phase of the signal.

4. Effect of Probe Compensation

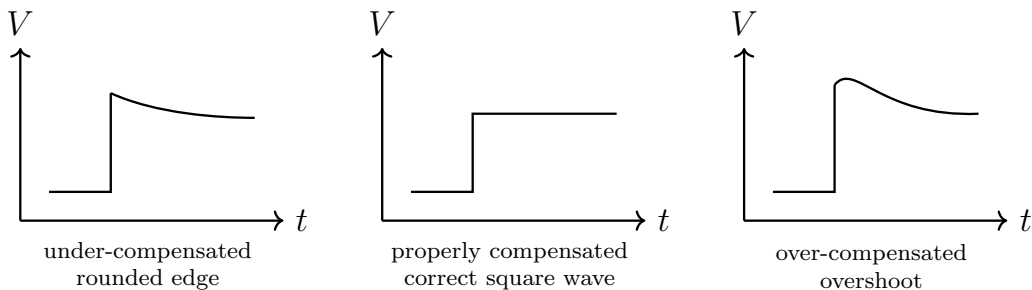


Figure 7: Effect of probe compensation on a displayed square wave.

5. Final Answer

Thus, a matched probe is necessary in a CRO to avoid frequency distortion, reduce loading effect, obtain accurate waveform display and make the probe-CRO combination act as a frequency-independent voltage divider.

The condition for matched probe is

$$R_p C_p = R_s C_s$$

Solution to Question 2. (a)

Problem Statement: What are the main advantages of switched capacitor circuit in integrated circuit technology?

Answer

A **switched capacitor circuit** is an electronic circuit in which a resistor is replaced by a capacitor and MOS switches. The capacitor is switched between two nodes at a high clock frequency. Due to this switching action, charge is transferred from one node to another. This charge transfer behaves like current flow through an equivalent resistor.

The equivalent resistance of a switched capacitor is

$$R_{eq} = \frac{1}{f_s C_s}$$

where,

$$f_s = \text{switching frequency}$$

and

$$C_s = \text{switched capacitor value}$$

Thus, by changing the clock frequency f_s , the equivalent resistance can be changed without changing any physical resistor.

Main Advantages

1. Suitable for IC fabrication:

In integrated circuits, fabrication of accurate resistors requires large chip area and is difficult. But capacitors and MOS switches can be fabricated more accurately and occupy less area. Therefore, switched capacitor circuits are highly suitable for integrated circuit technology.

2. Only capacitors and MOS switches are required:

A switched capacitor circuit replaces a resistor by using a capacitor and two MOS switches. Hence, large value resistors are avoided.

$$R_{\text{eq}} = \frac{1}{f_s C_s}$$

3. Large equivalent resistance can be obtained:

Very large resistances can be simulated using small capacitors and moderate clock frequencies. For example, a small capacitor switched at a suitable frequency can behave like a large resistor.

4. Frequency response can be controlled by clock frequency:

Since

$$R_{\text{eq}} = \frac{1}{f_s C_s}$$

the effective resistance depends on the switching frequency. Therefore, the filter characteristics can be tuned electronically by changing the clock frequency.

5. Better accuracy in filter design:

In IC technology, capacitor ratios can be made more accurately than resistor values. Hence, switched capacitor filters can provide accurate time constants and accurate filter response.

6. Reduced chip area:

Large resistors occupy large area in an integrated circuit. Switched capacitor circuits replace these large resistors by smaller capacitors and MOS switches. So, chip area is reduced.

7. Easy realization of active filters:

Switched capacitor circuits are very useful for realizing active filters such as low-pass, high-pass, band-pass and band-stop filters in integrated circuit form.

Final Answer

Thus, the main advantages of switched capacitor circuits in IC technology are compact size, good accuracy, electronic tunability, easy fabrication using MOS switches and capacitors, and the ability to simulate large resistances without using actual large resistors.

Solution to Question 2. (b)

Problem Statement: Find the equivalent switched capacitor circuit of the circuit shown. The switching frequency is 100 kHz. The value of $R = 10\text{ k}\Omega$, $C_1 = 1\text{ nF}$ and $C_2 = 10\text{ pF}$.

Answer

The given circuit contains two cascaded inverting active filter stages. Each stage has:

input resistor R

and

feedback network $R \parallel C$

In the switched capacitor equivalent circuit, each resistor R is replaced by a switched capacitor. The original capacitors C_1 and C_2 remain unchanged.

1. Formula for Equivalent Switched Capacitor

For a switched capacitor circuit,

$$R_{\text{eq}} = \frac{1}{f_s C_s}$$

Therefore,

$$C_s = \frac{1}{R_{\text{eq}} f_s}$$

Here,

$$R_{\text{eq}} = R = 10\text{ k}\Omega$$

$$R = 10 \times 10^3\ \Omega$$

and

$$f_s = 100\text{ kHz}$$

$$f_s = 100 \times 10^3\text{ Hz}$$

Therefore,

$$C_s = \frac{1}{(10 \times 10^3)(100 \times 10^3)}$$

$$C_s = \frac{1}{10^9}$$

$$C_s = 10^{-9}\text{ F}$$

$$C_s = 1 \text{ nF}$$

Thus, every $10 \text{ k}\Omega$ resistor in the given circuit is replaced by a switched capacitor of value

$$1 \text{ nF}$$

2. Values Used in the Equivalent Circuit

The given circuit has four resistors of value $R = 10 \text{ k}\Omega$. Therefore, each of them is replaced by a 1 nF switched capacitor.

$$R_{\text{input},1} \rightarrow C_{S1} = 1 \text{ nF}$$

$$R_{\text{feedback},1} \rightarrow C_{S2} = 1 \text{ nF}$$

$$R_{\text{input},2} \rightarrow C_{S3} = 1 \text{ nF}$$

$$R_{\text{feedback},2} \rightarrow C_{S4} = 1 \text{ nF}$$

The original capacitors are unchanged:

$$C_1 = 1 \text{ nF}$$

$$C_2 = 10 \text{ pF}$$

3. Basic Replacement Rule

Each resistor R is replaced by a switched capacitor block as shown below.

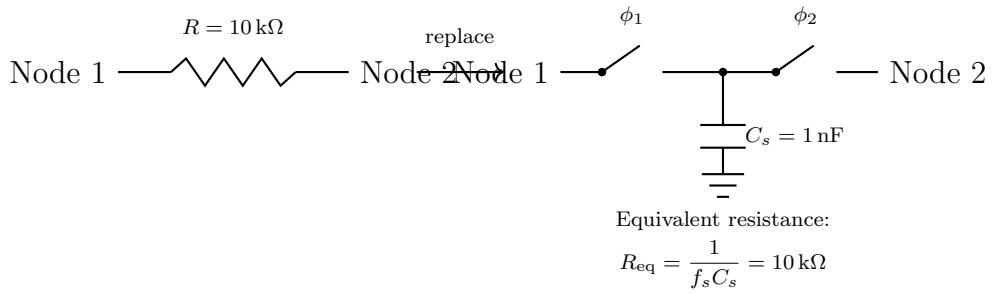


Figure 8: Replacement of a resistor by a switched capacitor block.

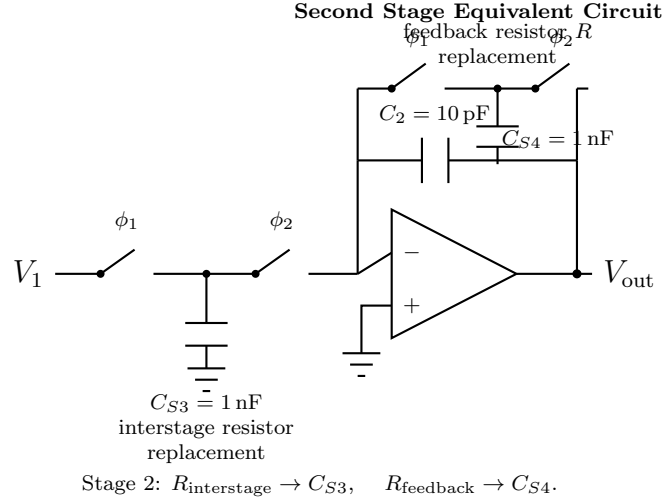


Figure 9: Second stage switched capacitor equivalent circuit.

5. Final Answer

Since,

$$R = 10 \text{ k}\Omega$$

and

$$f_s = 100 \text{ kHz}$$

the switched capacitor value required to replace each resistor is

$$C_s = 1 \text{ nF}$$

Therefore, the equivalent switched capacitor circuit is obtained by replacing every $10 \text{ k}\Omega$ resistor by a 1 nF switched capacitor block, while keeping the original capacitors unchanged:

$$C_1 = 1 \text{ nF}$$

$$C_2 = 10 \text{ pF}$$

Thus,

$$C_{S1} = C_{S2} = C_{S3} = C_{S4} = 1 \text{ nF}$$

Solution to Question 2. (c)

Problem Statement: How can you realize a second order high-pass filter using switched capacitor circuits?

Answer

A **switched capacitor circuit** replaces a physical resistor by a capacitor which is periodically switched between two nodes by two non-overlapping clock phases. This is highly useful in integrated-circuit filter design because accurate large-value resistors are difficult to fabricate on chip, whereas capacitor ratios and MOS switches are easier to realize.

For a switched capacitor of value C_s , switched at frequency f_s , the equivalent resistance is

$$R_{eq} = \frac{1}{f_s C_s}.$$

Therefore, a continuous-time active RC high-pass filter can be converted into a switched-capacitor high-pass filter by replacing each resistor by a switched capacitor block of suitable value.

1. Standard Second Order High-Pass Response

The general second order high-pass transfer function is

$$H_{HP}(s) = \frac{K s^2}{s^2 + \left(\frac{\omega_0}{Q}\right) s + \omega_0^2},$$

where

K = pass-band gain, ω_0 = cut-off or natural angular frequency,

Q = quality factor.

At low frequency ($s \rightarrow 0$),

$$H_{HP}(0) = 0.$$

Thus DC and low-frequency components are blocked. At high frequency ($s \rightarrow \infty$),

$$H_{HP}(s) \rightarrow K.$$

Thus high-frequency components are passed with gain K .

2. Resistor to Switched Capacitor Replacement

Every resistor in the RC high-pass topology can be replaced by the following equivalent switched capacitor block.

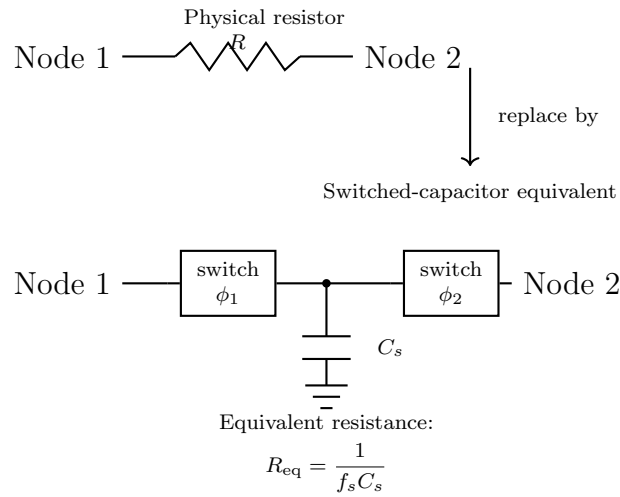
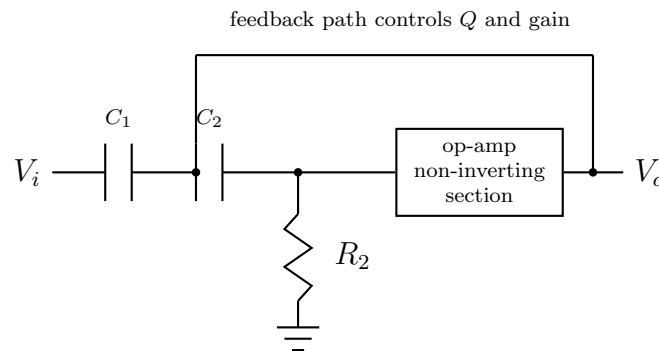


Figure 1: Basic switched-capacitor replacement of a resistor.

3. Active RC Second Order High-Pass Section

A second order high-pass filter may be realized by a Sallen-Key high-pass topology. A simplified exam-style block representation is shown below. The capacitors form the frequency-dependent input path and the resistors provide the reference paths. In switched-capacitor IC realization, the resistors are replaced by switched capacitor blocks.



High-pass section: capacitors are in the input path, so DC is blocked.

Figure 2: Conceptual second order active high-pass filter before switched-capacitor conversion.

4. Switched-Capacitor Realization

In IC form, each resistor R_i is replaced by an equivalent switched capacitor C_{Si} , where

$$C_{Si} = \frac{1}{R_i f_s}$$

The active element remains an op-amp, and the clock phases ϕ_1 and ϕ_2 must be non-overlapping.

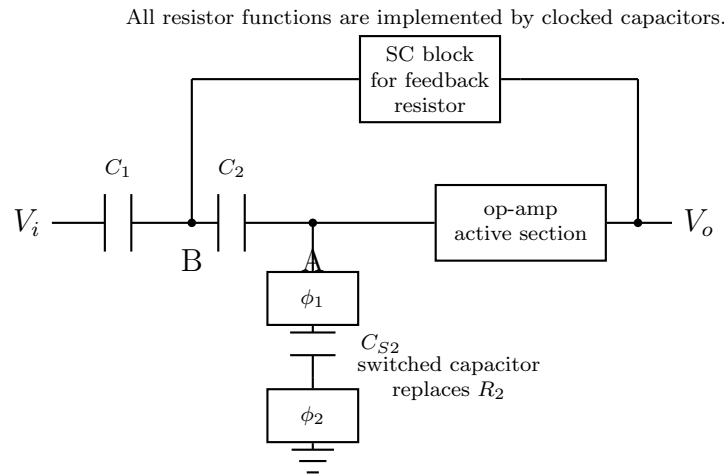


Figure 3: Switched-capacitor realization idea for a second order high-pass filter.

5. Design Steps

Step 1: Choose the required ω_0 , Q , and gain K from the high-pass specification.

Step 2: Select a suitable active RC high-pass topology such as Sallen-Key high-pass or state-variable high-pass.

Step 3: Identify every resistor R_i in the topology.

Step 4: Replace each resistor by a switched capacitor C_{Si} using

$$C_{Si} = \frac{1}{R_i f_s}.$$

Step 5: Drive all switched capacitor blocks with non-overlapping clock phases ϕ_1 and ϕ_2 .

Step 6: Keep the designed filter capacitors or capacitor ratios unchanged according to the selected topology.

Final Answer

A second order high-pass filter can be realized in switched capacitor form by starting from a second order active RC high-pass circuit and replacing each resistor by a switched capacitor equivalent:

$$R_i = \frac{1}{f_s C_{Si}} \implies C_{Si} = \frac{1}{R_i f_s}.$$

The resulting circuit is very suitable for IC technology because the cut-off frequency depends on capacitor ratios and clock frequency rather than on absolute resistor values.

Solution to Question 2. (OR) (a)

Problem Statement: List the basic building blocks of PLL. Explain the role of low pass filter and function of VCO in it.

Answer

A **Phase Locked Loop (PLL)** is a closed loop feedback system in which the output phase and frequency of a voltage controlled oscillator are locked to the phase and frequency of an input reference signal.

1. Basic Building Blocks of PLL

The basic blocks of a PLL are:

1. Phase detector or phase comparator.
2. Low pass filter or loop filter.
3. Voltage controlled oscillator (VCO).
4. Feedback path, sometimes including a frequency divider.

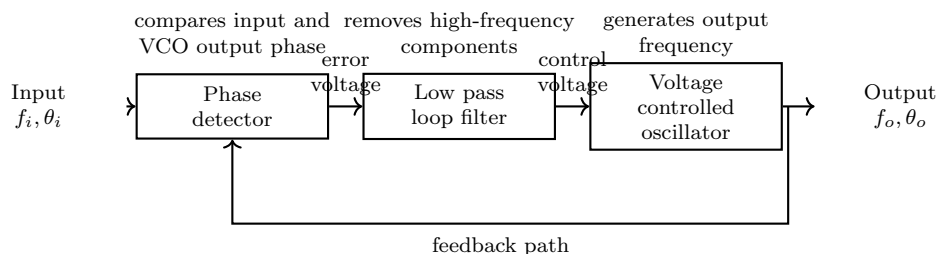


Figure 4: Basic building blocks of a phase locked loop.

2. Role of Low Pass Filter

The phase detector output generally contains a DC or slowly varying component proportional to phase error and high-frequency components due to multiplication/comparison action. The **low pass filter** performs the following functions:

1. It removes the unwanted high-frequency components from the phase detector output.
2. It produces a smooth DC or slowly varying control voltage.
3. It determines important dynamic properties such as capture range, lock range, stability and transient response.
4. It prevents rapid unwanted variation of VCO frequency.

If the phase detector output is $V_e(t)$, then the filter output is

$$V_d(s) = F(s)V_e(s),$$

where $F(s)$ is the loop filter transfer function.

3. Function of VCO

The **Voltage Controlled Oscillator (VCO)** is an oscillator whose output frequency depends on the applied control voltage. Its characteristic is

$$\omega_o(t) = \omega_{FR} + K_v V_d(t),$$

where

ω_{FR} = free-running angular frequency,

K_v = VCO sensitivity in rad/sec/volt,

$V_d(t)$ = control voltage from loop filter.

When the PLL is locked, the VCO frequency becomes equal to the input signal frequency. If the input frequency changes slightly, the phase error changes, the loop filter output changes, and the VCO frequency is adjusted until lock is restored.

Final Answer

The basic PLL blocks are phase detector, low pass filter, VCO and feedback path. The low pass filter smooths the phase detector output and controls loop dynamics. The VCO generates the PLL output frequency and varies it according to the control voltage so that the output stays locked to the input.

Solution to Question 2. (OR) (b)

Problem Statement: A PLL based frequency translator employs a divide by N counter. The reference frequency is 10 kHz. Find N for translating frequency from 50 kHz to 130 kHz. Illustrate with suitable schematic diagram.

Answer

A PLL frequency synthesizer or frequency translator uses a programmable divide-by- N counter in the feedback path. The phase detector compares the reference frequency f_{ref} with the divided output frequency f_o/N .

Under locked condition,

$$\frac{f_o}{N} = f_{ref}.$$

Therefore,

$$f_o = N f_{ref}.$$

1. Block Diagram

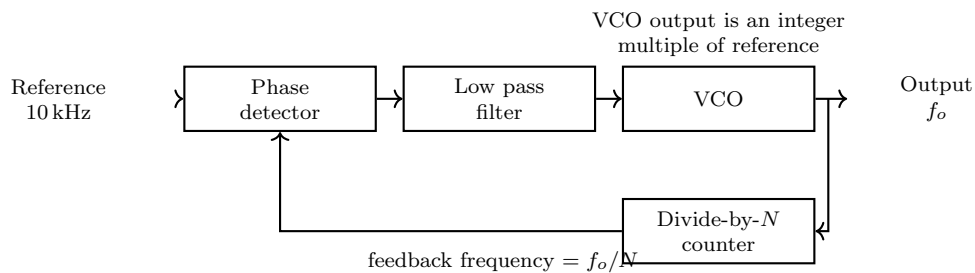


Figure 5: PLL based frequency translator using divide-by- N counter.

2. Calculation of N

Given reference frequency,

$$f_{ref} = 10 \text{ kHz.}$$

The required output range is

$$50 \text{ kHz} \leq f_o \leq 130 \text{ kHz.}$$

Using

$$f_o = N f_{ref},$$

we get

$$N = \frac{f_o}{f_{ref}}.$$

For the lower frequency,

$$N_{min} = \frac{50 \text{ kHz}}{10 \text{ kHz}} = 5.$$

For the upper frequency,

$$N_{max} = \frac{130 \text{ kHz}}{10 \text{ kHz}} = 13.$$

Therefore,

$$N = 5, 6, 7, 8, 9, 10, 11, 12, 13.$$

3. Frequency Table

N	Output frequency $f_o = N \times 10 \text{ kHz}$
5	50 kHz
6	60 kHz
7	70 kHz
8	80 kHz
9	90 kHz
10	100 kHz
11	110 kHz
12	120 kHz
13	130 kHz

Final Answer

To translate the frequency from 50 kHz to 130 kHz using 10 kHz reference, the divide counter must be varied from

$$N = 5 \text{ to } 13.$$

Solution to Question 2. (OR) (c)

Problem Statement: Explain lock range and capture range of a PLL.

Answer

A PLL can operate in three modes: free-running mode, capture mode and lock mode. Two important frequency ranges associated with PLL operation are **capture range** and **lock range**.

1. Capture Range

The **capture range** is the range of input signal frequencies around the free-running frequency of the VCO for which an initially unlocked PLL can acquire lock.

If the input frequency is within the capture range, the PLL can pull the VCO frequency towards the input frequency and finally lock it.

Thus,

Capture range is the acquisition range of PLL.

The capture range depends mainly on the low pass filter characteristics. It is usually narrower than lock range.

2. Lock Range

The **lock range** is the range of input frequencies around the center frequency for which an already locked PLL remains locked.

Once the PLL is locked, it can track changes in input frequency over a wider range. If the input frequency goes beyond this range, the PLL loses lock.

Thus,

Lock range is the tracking range of PLL.

3. Diagram

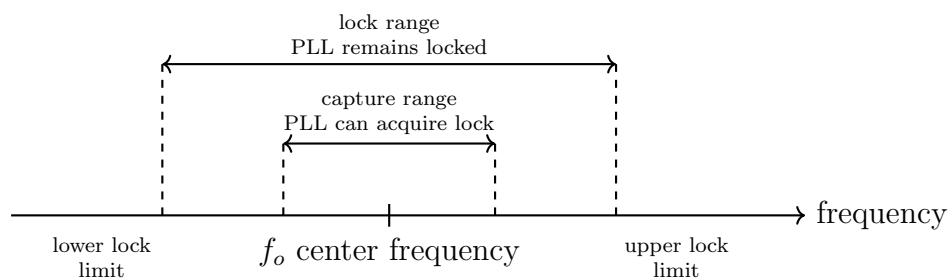


Figure 6: Capture range and lock range of a PLL.

4. Difference Between Lock Range and Capture Range

Capture Range	Lock Range
Range over which PLL can acquire lock from initially unlocked state.	Range over which PLL remains locked after lock has already been achieved.
Usually smaller.	Usually larger.
Depends strongly on loop filter.	Depends on VCO tuning range and loop gain.
Associated with pull-in process.	Associated with tracking process.

Final Answer

Capture range: range over which an unlocked PLL can acquire lock.

Lock range: range over which an already locked PLL remains locked.

Solution to Question 3. (a)

Problem Statement: The step size of a 9-bit D/A converter is 10.3 mV. If all zeros represent 0 V, what output is produced for the input 101101111? What are the limitations of the weighted resistor type D/A converter?

Answer

The step size of a DAC is the output voltage change corresponding to a change of one least significant bit. It is also called 1 LSB.

Given,

$$1 \text{ LSB} = 10.3 \text{ mV} = 0.0103 \text{ V}.$$

All zeros represent zero volt, therefore the output is

$$V_o = D(\text{step size}),$$

where D is the decimal equivalent of the digital input.

1. Binary to Decimal Conversion

Given digital input:

$$101101111_2.$$

For a 9-bit number,

$$101101111_2 = 1 \times 2^8 + 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0.$$

Thus,

$$D = 256 + 0 + 64 + 32 + 0 + 8 + 4 + 2 + 1.$$

$$D = 367.$$

2. Output Voltage

$$V_o = D \times 1 \text{ LSB}.$$

$$V_o = 367 \times 10.3 \text{ mV}.$$

$$V_o = 3780.1 \text{ mV}.$$

$$\boxed{V_o = 3.7801 \text{ V}}.$$

3. Limitations of Weighted Resistor DAC

A binary weighted resistor DAC uses resistors weighted as $R, 2R, 4R, 8R, \dots$. The current contributed by each bit is proportional to its binary weight.

- 1. Wide range of resistor values:** For high-resolution DACs, very large and very small resistor values are required. This is difficult to fabricate accurately.

2. **Poor accuracy for many bits:** Resistor tolerance causes incorrect bit weights and conversion error.
3. **Difficult IC fabrication:** Large resistor ratios occupy large chip area and are not suitable for integrated circuits.
4. **Switch resistance error:** Practical switches have finite ON resistance, which changes the intended binary current weights.
5. **Loading and settling problems:** Different switch positions and resistor values affect output resistance and settling time.

Final Answer

$$101101111_2 = 367_{10}$$

$$V_o = 367 \times 10.3 \text{ mV} = 3.7801 \text{ V}.$$

The weighted resistor DAC is limited by the need for a wide range of accurate resistor values and is therefore not preferred for high-resolution IC DACs.

Solution to Question 3. (b)

Problem Statement: The analog values corresponding to digital signals 000 and 001 are 0 V and 0.625 V respectively. Plot the transfer function curve of this DAC with and without an offset of 0.5 LSB.

Answer

The given digital inputs have 3 bits. Therefore, the possible codes are

$$000, 001, 010, 011, 100, 101, 110, 111.$$

The output corresponding to 000 is 0 V, and the output corresponding to 001 is 0.625 V. Therefore,

$$1 \text{ LSB} = 0.625 \text{ V}.$$

1. Without 0.5 LSB Offset

Without offset, the output voltage is

$$V_o = D \times 0.625 \text{ V},$$

where D is the decimal equivalent of the input code.

Digital code	Decimal D	Output without offset
000	0	0
001	1	0.625 V
010	2	1.250 V
011	3	1.875 V
100	4	2.500 V
101	5	3.125 V
110	6	3.750 V
111	7	4.375 V

2. With 0.5 LSB Offset

Half LSB is

$$0.5 \text{ LSB} = \frac{0.625}{2} = 0.3125 \text{ V}.$$

With offset,

$$V_o = (D + 0.5) \times 0.625 \text{ V}.$$

Digital code	Decimal D	Output with 0.5 LSB offset
000	0	0.3125 V
001	1	0.9375 V
010	2	1.5625 V
011	3	2.1875 V
100	4	2.8125 V
101	5	3.4375 V
110	6	4.0625 V
111	7	4.6875 V

3. Transfer Function Plot

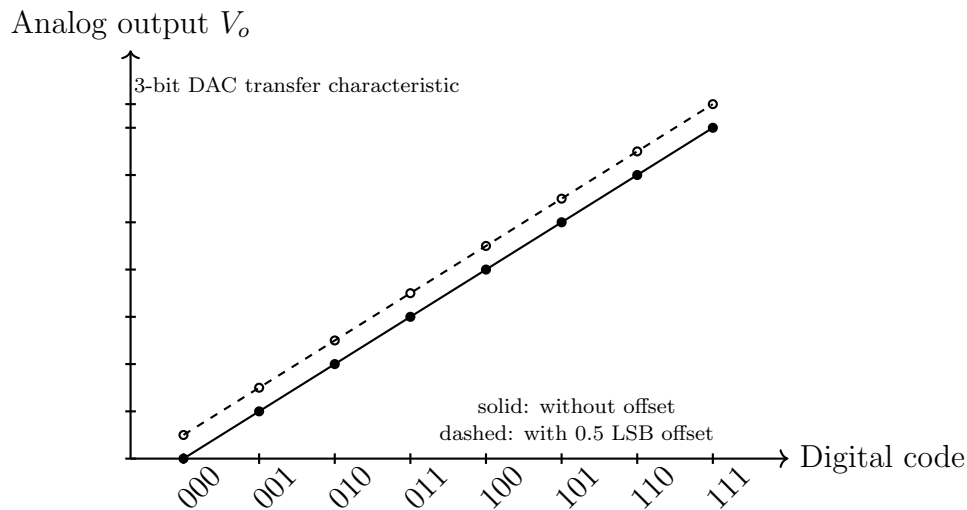


Figure 7: DAC transfer characteristic with and without 0.5 LSB offset.

Final Answer

$$1 \text{ LSB} = 0.625 \text{ V}$$

Without offset,

$$V_o = D(0.625) \text{ V}.$$

With 0.5 LSB offset,

$$V_o = (D + 0.5)(0.625) \text{ V}.$$

Solution to Question 3. (c)

Problem Statement: Define settling time and monotonicity with reference to D/A converter.

Answer

Two important dynamic and static specifications of a DAC are **settling time** and **monotonicity**.

1. Settling Time

Settling time is the time required by the DAC output to reach and remain within a specified error band around its final value after a change in digital input code.

Usually, the error band is specified as

$$\pm \frac{1}{2} \text{LSB}$$

or as a percentage of full-scale output.

If the digital input suddenly changes from one code to another, the analog output does not change instantaneously. It takes some time due to switching delay, op-amp slew rate, capacitance and internal circuit dynamics.

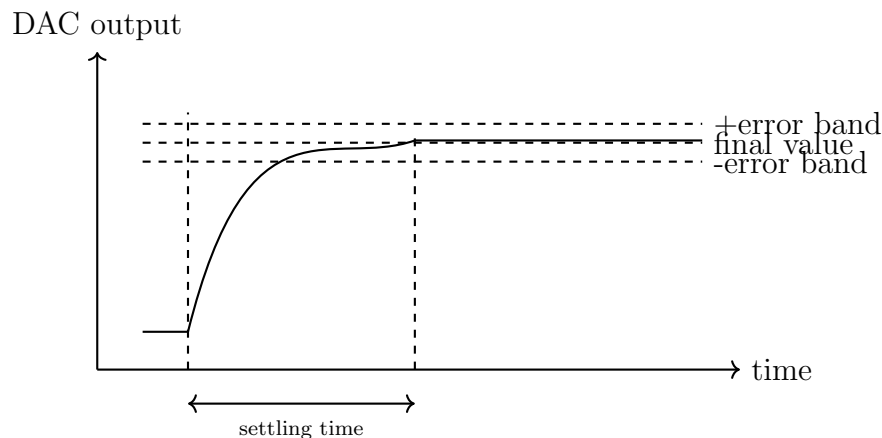


Figure 8: Settling time of a DAC output.

2. Monotonicity

A DAC is said to be **monotonic** if its analog output never decreases when the digital input code is increased. In other words, for increasing input code, the output must either increase or remain the same.

Mathematically, if

$$D_2 > D_1,$$

then a monotonic DAC must satisfy

$$V_o(D_2) \geq V_o(D_1).$$

A non-monotonic DAC is undesirable because an increase in digital input may produce a decrease in analog output. This can cause serious errors in control systems and feedback applications.

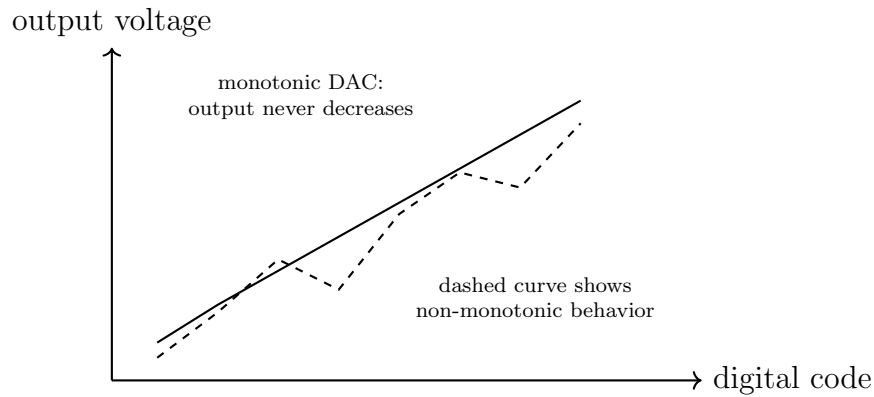


Figure 9: Monotonic and non-monotonic DAC characteristics.

Final Answer

Settling time is the time required by DAC output to reach its final value within a specified error band.

Monotonicity means DAC output never decreases when input code is increased.

Solution to Question 3. (OR) (a)

Problem Statement: For the ADC circuit shown in the figure, what will be the digital output? Give reasons in support of your answer.

Answer

The given circuit is a **counter type ADC**. It contains a comparator, a 4-bit counter, a 4-bit DAC and an AND gate controlling the clock. The DAC output is compared with the analog input voltage.

From the figure,

$$V_{in} = 6.2 \text{ V}, \quad V_{ref} = 8 \text{ V}.$$

The DAC is a 4-bit DAC, so

$$n = 4.$$

1. Clean Functional Diagram

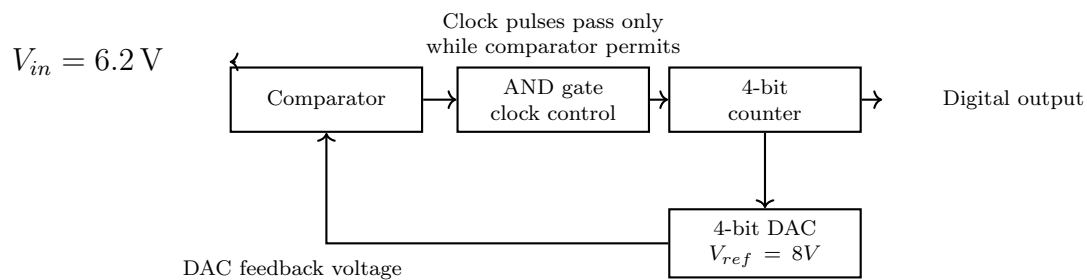


Figure 10: Counter type ADC represented from the given figure.

2. Step Size of 4-bit DAC

For an ideal unipolar 4-bit DAC,

$$\Delta = \frac{V_{ref}}{2^n}.$$

Substituting,

$$\Delta = \frac{8}{2^4} = \frac{8}{16} = 0.5 \text{ V}.$$

Thus each count increases the DAC output by

$$\boxed{0.5 \text{ V}}.$$

3. Determine Required Count

The counter will count until the DAC output just reaches or exceeds the input voltage.

For count D ,

$$V_{DAC} = D \times 0.5 \text{ V}.$$

We need

$$V_{DAC} \geq 6.2 \text{ V}.$$

Thus,

$$D \times 0.5 \geq 6.2.$$

$$D \geq 12.4.$$

The next integer count is

$$D = 13.$$

For $D = 12$,

$$V_{DAC} = 12 \times 0.5 = 6.0 \text{ V} < 6.2 \text{ V}.$$

For $D = 13$,

$$V_{DAC} = 13 \times 0.5 = 6.5 \text{ V} > 6.2 \text{ V}.$$

Hence the counter stops at $D = 13$.

4. Decimal to Binary Conversion

$$13_{10} = 1101_2.$$

Final Answer

Digital output = 1101

The output is 1101 because the 4-bit DAC output first exceeds 6.2 V at count 13, where $V_{DAC} = 6.5 \text{ V}$.

Solution to Question 3. (OR) (b)

Problem Statement: What is the function of sample and hold amplifier circuit in ADC?

Answer

A **sample and hold amplifier** is used before an ADC to sample the instantaneous value of an analog input signal and hold that value constant during the conversion interval.

1. Need of Sample and Hold Circuit

An ADC requires a finite time to convert an analog voltage into a digital code. If the input voltage changes during this conversion time, the ADC may produce an incorrect digital output. The sample and hold circuit prevents this error by keeping the input to the ADC constant during conversion.

2. Basic Diagram

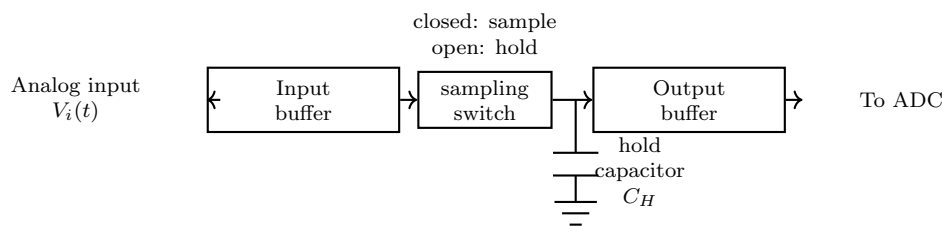


Figure 11: Sample and hold amplifier used before an ADC.

3. Operation

1. **Sample mode:** The switch is closed. The hold capacitor charges to the instantaneous input voltage.
2. **Hold mode:** The switch is opened. The capacitor retains the sampled voltage.
3. **Conversion mode:** The ADC converts this constant voltage into a digital code.

4. Main Functions

1. Holds the input voltage constant during ADC conversion.
2. Improves conversion accuracy for time-varying signals.
3. Prevents aperture error caused by input signal variation.
4. Allows the ADC to work properly with fast-changing analog signals.

Final Answer

The sample and hold amplifier samples the analog signal and holds it constant during ADC conversion.

Solution to Question 3. (OR) (c)

Problem Statement: Explain gain and linearity errors of an ADC.

Answer

An ADC converts an analog input voltage into a digital output code. Ideally, the transition points between output codes lie on a straight line. In practical ADCs, deviations occur due to offset, gain error, component mismatch and comparator imperfections.

1. Gain Error

Gain error is the error in the slope of the actual ADC transfer characteristic compared with the ideal transfer characteristic, after offset error has been removed.

If the actual full-scale transition occurs at a different input voltage than the ideal full-scale transition, the ADC has gain error.

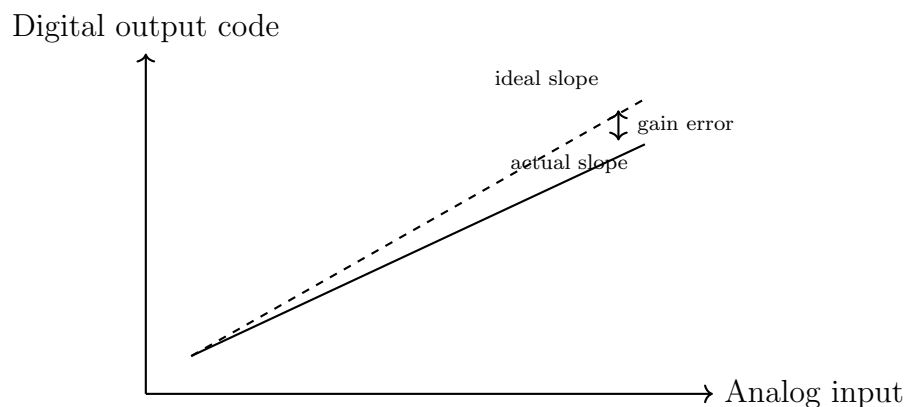


Figure 12: Gain error in ADC transfer characteristic.

2. Linearity Error

Linearity error is the deviation of the actual ADC transition points from the ideal straight line after offset and gain errors have been corrected.

It is generally expressed in LSB. The two common linearity errors are:

1. Integral non-linearity (INL).
2. Differential non-linearity (DNL).

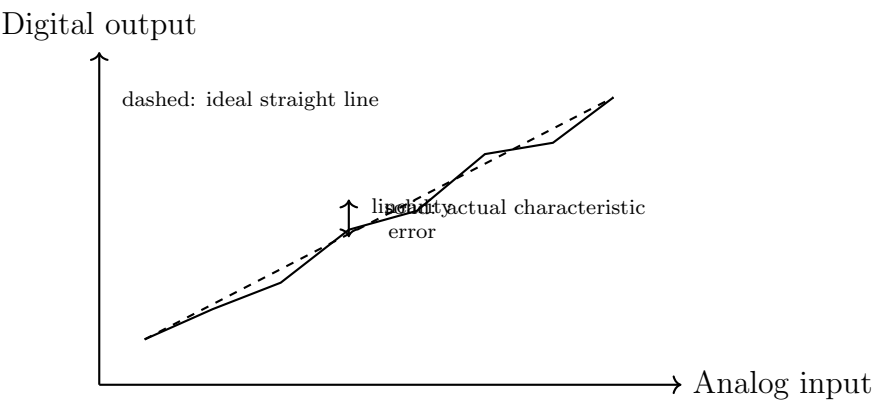


Figure 13: Linearity error in ADC transfer characteristic.

3. Comparison

Gain Error	Linearity Error
Error in slope of transfer characteristic. Visible near full-scale output.	Deviation from straight-line behavior. May occur anywhere in the conversion range.
Correctable by calibration of gain.	More difficult to correct; depends on internal matching.
Measured after offset error is removed.	Measured after offset and gain errors are removed.

Final Answer

Gain error is slope error of ADC transfer curve after offset correction.

Linearity error is deviation of actual transition points from the ideal straight line.

Solution to Question 4. (a)

Problem Statement: Mention the differences between the properties of Butterworth and Chebyshev filters. Explain why the -3 dB cut-off frequency of a normalized Chebyshev filter is greater than 1 rad/sec.

Answer

Butterworth and Chebyshev filters are two important approximation techniques used in filter design.

1. Difference Between Butterworth and Chebyshev Filters

Butterworth Filter	Chebyshev Filter
It has maximally flat magnitude response in the pass band.	It has equiripple response in the pass band.
No ripple in pass band.	Ripple is present in pass band.
Roll-off is slower compared with Chebyshev filter of same order.	Roll-off is sharper than Butterworth filter of same order.
Poles lie on a circle in the normalized s -plane.	Poles lie on an ellipse in the normalized s -plane.
Magnitude response is monotonic.	Magnitude response oscillates in pass band.
Used when flat pass-band response is most important.	Used when sharp transition is required.

2. Chebyshev Magnitude Response

For a normalized Chebyshev low-pass filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)},$$

where ϵ is the ripple factor and $T_n(\omega)$ is the Chebyshev polynomial.

For a normalized Chebyshev filter, $\omega = 1$ rad/sec represents the end of pass-band ripple region, not necessarily the -3 dB frequency.

At $\omega = 1$,

$$T_n(1) = 1.$$

Therefore,

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}.$$

For normal Chebyshev filters, the pass-band ripple is less than 3 dB, hence

$$\epsilon < 1.$$

Thus,

$$1 + \epsilon^2 < 2.$$

Therefore,

$$\frac{1}{1 + \epsilon^2} > \frac{1}{2}.$$

But -3 dB means half-power point:

$$|H(j\omega_{3dB})|^2 = \frac{1}{2}.$$

Since at $\omega = 1$ the magnitude squared is still greater than $1/2$, the frequency must increase beyond 1 rad/sec to reach the -3 dB point.

3. Formula for -3 dB Frequency

At -3 dB,

$$\frac{1}{1 + \epsilon^2 T_n^2(\omega_{3dB})} = \frac{1}{2}.$$

Hence,

$$T_n(\omega_{3dB}) = \frac{1}{\epsilon}.$$

For $\omega > 1$,

$$T_n(\omega) = \cosh[n \cosh^{-1}(\omega)].$$

Therefore,

$$\omega_{3dB} = \cosh \left[\frac{1}{n} \cosh^{-1} \left(\frac{1}{\epsilon} \right) \right].$$

Since $1/\epsilon > 1$,

$$\boxed{\omega_{3dB} > 1 \text{ rad/sec}}.$$

4. Diagram

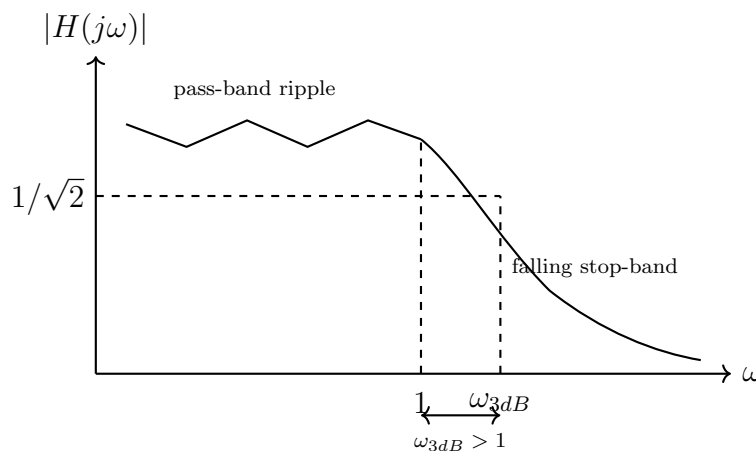


Figure 14: Normalized Chebyshev filter: the -3 dB point occurs after 1 rad/sec.

Final Answer

Butterworth response is flat in the pass band, while Chebyshev response has pass-band ripple but sharper roll-off. In a normalized Chebyshev filter, $\omega = 1 \text{ rad/sec}$ is the end of pass-band ripple, and the half-power point occurs later. Hence

$$\omega_{3dB} > 1 \text{ rad/sec}.$$

Solution to Question 4. (b)

Problem Statement: Draw the circuit diagram of Sallen-Key second order band-pass filter circuit. Write down the expression for transfer function of a band-pass filter in terms of center frequency ω_0 , bandwidth B and passband gain 0 dB. How can this filter be converted to band-stop filter without altering the specification?

Answer

A **band-pass filter** passes a band of frequencies around a center frequency ω_0 and attenuates frequencies below and above this band. A second order band-pass filter has two poles and one zero at the origin.

1. Sallen-Key Band-Pass Circuit

A clean representation of a Sallen-Key second order band-pass circuit is shown below. Labels are placed away from the wires for clarity.

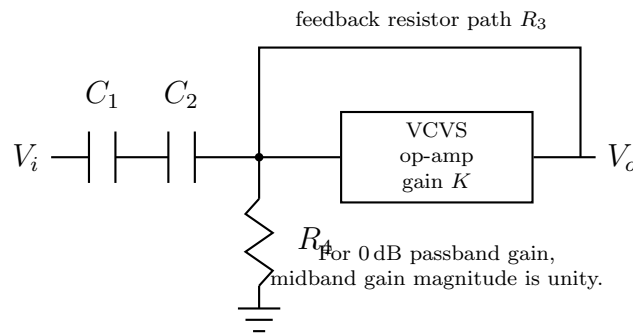


Figure 15: Sallen-Key second order band-pass filter circuit representation.

2. Transfer Function in Terms of ω_0 , B , and Passband Gain

The standard second order band-pass transfer function is

$$H_{BP}(s) = \frac{A_m B s}{s^2 + B s + \omega_0^2},$$

where

ω_0 = center angular frequency,

B = bandwidth,

A_m = passband or midband gain.

For 0 dB passband gain,

$$A_m = 1.$$

Therefore,

$$H_{BP}(s) = \frac{B s}{s^2 + B s + \omega_0^2}.$$

Quality factor is

$$Q = \frac{\omega_0}{B}.$$

Thus, the same transfer function can also be written as

$$H_{BP}(s) = \frac{\left(\frac{\omega_0}{Q}\right)s}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}.$$

3. Verification of Midband Gain

At center frequency,

$$s = j\omega_0.$$

Then

$$(j\omega_0)^2 + \omega_0^2 = 0.$$

Therefore,

$$H_{BP}(j\omega_0) = \frac{Bj\omega_0}{Bj\omega_0} = 1.$$

Hence,

$$\boxed{|H_{BP}(j\omega_0)| = 1 = 0 \text{ dB}}.$$

4. Conversion to Band-Stop Filter

A band-stop filter with the same center frequency and bandwidth can be obtained by subtracting the band-pass output from a direct unity path:

$$H_{BS}(s) = 1 - H_{BP}(s).$$

Substituting,

$$H_{BS}(s) = 1 - \frac{Bs}{s^2 + Bs + \omega_0^2}.$$

$$H_{BS}(s) = \frac{s^2 + Bs + \omega_0^2 - Bs}{s^2 + Bs + \omega_0^2}.$$

Therefore,

$$\boxed{H_{BS}(s) = \frac{s^2 + \omega_0^2}{s^2 + Bs + \omega_0^2}}.$$

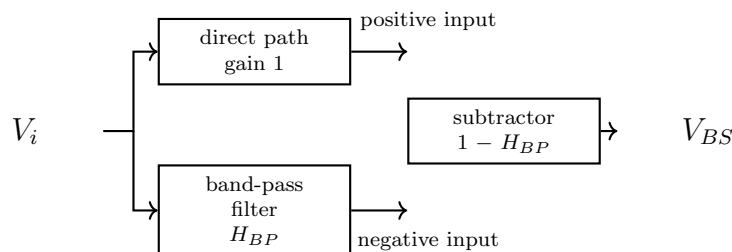


Figure 16: Conversion of band-pass filter to band-stop filter without changing ω_0 and B .

Final Answer

For 0 dB passband gain,

$$H_{BP}(s) = \frac{Bs}{s^2 + Bs + \omega_0^2}.$$

The corresponding band-stop response with the same specification is

$$H_{BS}(s) = \frac{s^2 + \omega_0^2}{s^2 + Bs + \omega_0^2}.$$

Solution to Question 4. (c)

Problem Statement: The filter circuit shown resembles which type of filter? Determine the transfer function and quality factor Q of this filter. Assume $R_A = 33\text{ k}\Omega$, $R_B = 10\text{ k}\Omega$, and $C_A = C_B = 100\text{ }\mu\text{F}$.

Answer

The given circuit consists of two buffered RC sections connected in cascade:

1. First section: series capacitor C_A and shunt resistor R_A . This is a high-pass section.
2. Second section: series resistor R_B and shunt capacitor C_B . This is a low-pass section.

A high-pass section followed by a low-pass section forms a **band-pass filter**.

1. Clean Circuit Representation

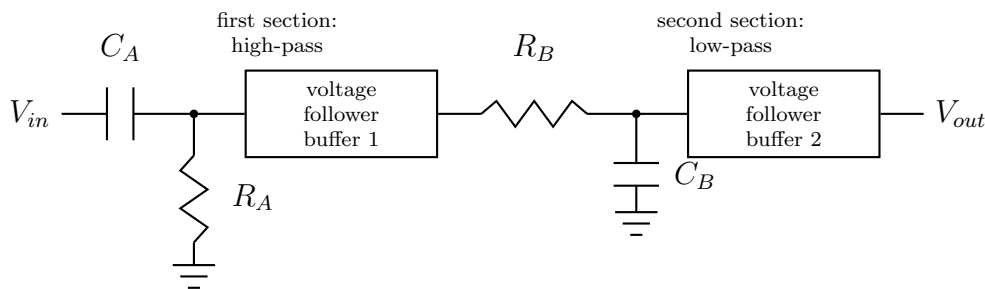


Figure 17: Given circuit redrawn as cascaded high-pass and low-pass buffered sections.

2. Transfer Function of First Section

The first section is a high-pass RC section:

$$H_A(s) = \frac{sR_AC_A}{1 + sR_AC_A}.$$

Let

$$\tau_A = R_AC_A.$$

Then

$$H_A(s) = \frac{s\tau_A}{1 + s\tau_A}.$$

3. Transfer Function of Second Section

The second section is a low-pass RC section:

$$H_B(s) = \frac{1}{1 + sR_BC_B}.$$

Let

$$\tau_B = R_B C_B.$$

Then

$$H_B(s) = \frac{1}{1 + s\tau_B}.$$

4. Overall Transfer Function

Since buffers isolate the two sections, the total transfer function is the product:

$$H(s) = H_A(s)H_B(s).$$

$$H(s) = \frac{s\tau_A}{(1 + s\tau_A)(1 + s\tau_B)}.$$

Expanding the denominator,

$$H(s) = \frac{s\tau_A}{1 + s(\tau_A + \tau_B) + s^2\tau_A\tau_B}.$$

Rearranging,

$$H(s) = \frac{s\tau_A}{s^2\tau_A\tau_B + s(\tau_A + \tau_B) + 1}.$$

Dividing numerator and denominator by $\tau_A\tau_B$,

$$H(s) = \frac{\frac{s}{\tau_B}}{s^2 + s\left(\frac{\tau_A + \tau_B}{\tau_A\tau_B}\right) + \frac{1}{\tau_A\tau_B}}.$$

This is a second order band-pass form:

$$H(s) = \frac{K_B s}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}.$$

Therefore,

$$\omega_0^2 = \frac{1}{\tau_A\tau_B},$$

so

$$\omega_0 = \frac{1}{\sqrt{\tau_A\tau_B}}.$$

Also,

$$\frac{\omega_0}{Q} = \frac{\tau_A + \tau_B}{\tau_A\tau_B}.$$

Thus,

$$Q = \frac{\sqrt{\tau_A\tau_B}}{\tau_A + \tau_B}.$$

5. Numerical Calculation

Given,

$$R_A = 33 \text{ k}\Omega = 33 \times 10^3 \Omega,$$

$$R_B = 10 \text{ k}\Omega = 10 \times 10^3 \Omega,$$

$$C_A = C_B = 100 \mu\text{F} = 100 \times 10^{-6} \text{ F}.$$

Then,

$$\tau_A = R_A C_A = (33 \times 10^3)(100 \times 10^{-6}) = 3.3 \text{ s}.$$

$$\tau_B = R_B C_B = (10 \times 10^3)(100 \times 10^{-6}) = 1.0 \text{ s}.$$

Therefore,

$$H(s) = \frac{s(3.3)}{(1 + 3.3s)(1 + s)}.$$

$$H(s) = \frac{3.3s}{3.3s^2 + 4.3s + 1}.$$

Dividing by 3.3,

$$H(s) = \frac{s}{s^2 + 1.303s + 0.303}.$$

Center frequency:

$$\omega_0 = \frac{1}{\sqrt{3.3 \times 1}} = 0.550 \text{ rad/sec}.$$

Quality factor:

$$Q = \frac{\sqrt{3.3 \times 1}}{3.3 + 1}.$$

$$Q = \frac{1.817}{4.3} = 0.422.$$

Final Answer

The circuit is a

buffered RC band-pass filter.

The transfer function is

$$H(s) = \frac{s}{s^2 + 1.303s + 0.303}.$$

The quality factor is

$$Q = 0.422.$$